

ASSIGNMENT -01

Variance and Bias (Diagram, overfit, underfit). For best fit model should we have low bias or high variance, low bias or low variance, high bias or high variance, low bias or high variance

Introduction

Machine Learning models aim to learn patterns from data and make accurate predictions on unseen data. However, building a perfect model is challenging. Two fundamental sources of error in predictive modeling are **Bias** and **Variance**. These concepts explain why a model may perform poorly and how to improve it.

Understanding bias and variance is essential for:

- Model evaluation
- Avoiding overfitting
- Avoiding underfitting
- Selecting the best-fit model

The trade-off between bias and variance determines model performance.

What is Bias?

Definition

Bias refers to the error introduced by approximating a real-world problem with a simplified model.

A high-bias model:

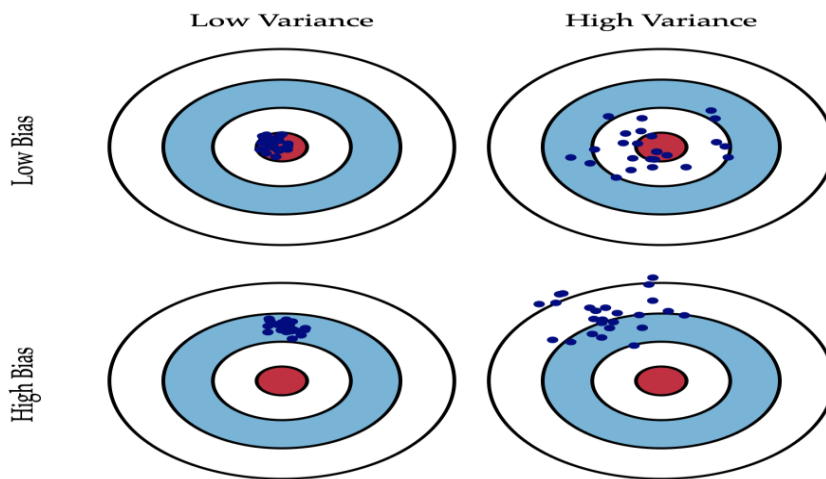
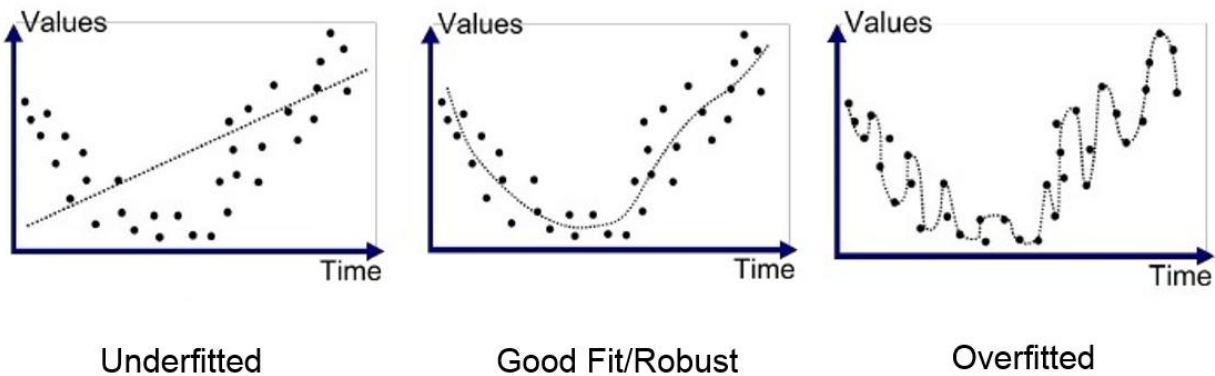
- Makes strong assumptions
- Is too simple
- Fails to capture underlying patterns
- Leads to **underfitting**

Mathematical Insight

Bias measures how far predictions are from the true values on average.

High bias → Model consistently wrong
Low bias → Model predictions closer to actual values

Diagram: High Bias (Underfitting)



In the above type of graph:

- Data points follow a curve.
- The model fits a straight line.
- The model ignores complexity.

What is Variance?

Definition

Variance refers to how much the model's predictions vary for different training datasets.

A high-variance model:

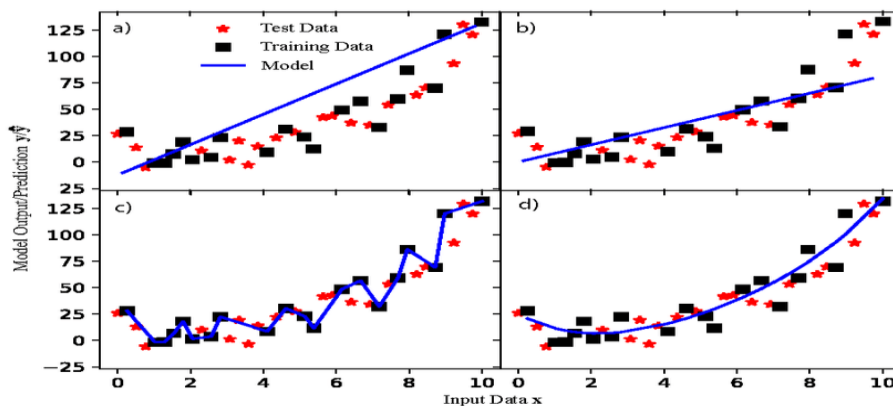
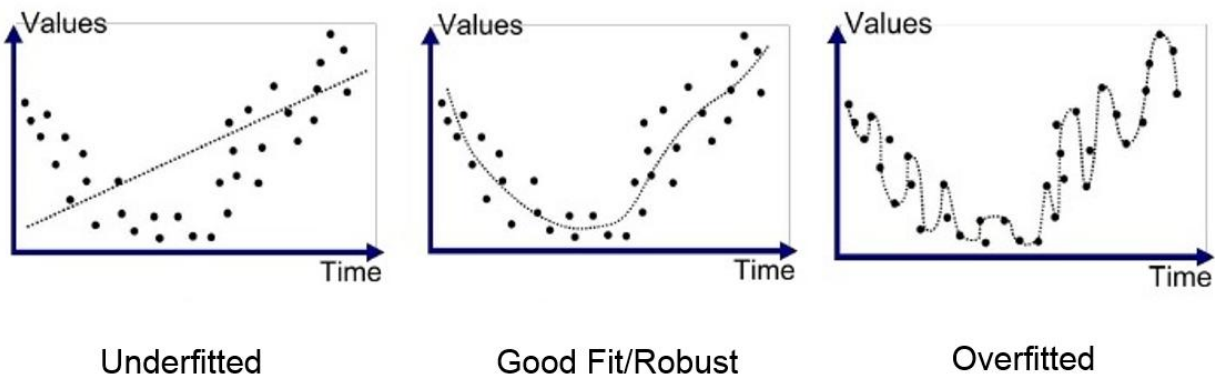
- Learns noise

- Is overly complex
- Changes significantly with small data changes
- Leads to **overfitting**

Low variance:

- Stable predictions
- Less sensitivity to data fluctuations

Diagram: High Variance (Overfitting)



In overfitting:

- The model fits every data point.
 - Curve becomes too complex.
 - Poor generalization on new data.
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Underfitting Explained

Underfitting occurs when:

- Model is too simple
- Cannot capture trends
- Performs poorly on both training and test data

Causes:

- Low model complexity
- Too few features
- Excessive regularization

Example:

Using a straight line for nonlinear data.

Effects:

- High bias
 - Low variance
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Overfitting Explained

Overfitting occurs when:

- Model memorizes training data
- Learns noise
- Performs well on training data
- Performs poorly on test data

Causes:

- Complex models
- Too many features
- Small dataset

Effects:

- Low bias

- High variance

Bias-Variance Tradeoff

The total error can be expressed as:

$$Total\ Error = Bias^2 + Variance + Irreducible\ Error$$

As model complexity increases:

- Bias decreases
- Variance increases

Graph Explanation:

- Left side: High bias (underfitting)
- Right side: High variance (overfitting)
- Middle: Optimal model

This is called the **Bias-Variance Tradeoff**.

Best Fit Model — What Should We Choose?

The correct answer is:

Low Bias and Low Variance

Let us analyze all options:

- | | | | |
|----|---|---|---------------|
| 1. | High Bias | & | High Variance |
| | Worst case — inaccurate and unstable. | | |
| 2. | High Bias | & | Low Variance |
| | Stable but consistently wrong (underfitting). | | |
| 3. | Low Bias | & | High Variance |
| | Accurate on training data but unstable (overfitting). | | |

4. Low Bias & Low Variance
Accurate and stable — Ideal model.

Thus, the best-fit model should have:

Low Bias and Low Variance

This ensures:

- Good training accuracy
 - Good test accuracy
 - Strong generalization
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Practical Example

Suppose we predict house prices.

- Simple linear model → Underfits → High bias
- 15-degree polynomial → Overfits → High variance
- 3rd-degree polynomial → Balanced → Best fit

The goal is not minimum training error, but minimum test error.

Techniques to Reduce Bias and Variance

To Reduce High Bias:

- Increase model complexity
- Add more features
- Reduce regularization

To Reduce High Variance:

- Use more training data
- Apply regularization (L1/L2)
- Pruning (for trees)
- Dropout (for neural networks)
- Cross-validation

Conclusion

Bias and Variance are two key components that determine machine learning model performance.

- **High Bias → Underfitting**
- **High Variance → Overfitting**
- **Balanced Model → Low Bias & Low Variance**

The goal of a good model is not extreme complexity or extreme simplicity but the right balance between bias and variance.

The Bias-Variance Tradeoff explains why increasing complexity does not always improve performance. The best-fit model lies at the point where total error is minimized.

Understanding these concepts helps in:

- Selecting proper algorithms
- Improving model accuracy
- Building reliable predictive systems

In summary:

For the best fit model, we should aim for LOW BIAS and LOW VARIANCE.